

Sec: XI_STU_IC_IIT
Time: 3 Hrs.

BCTM

Date: 20-05-2024
Max. Marks: 300

JEE MAIN 2022 Model

PHYSICS :

Section	Question Type	+Ve Marks	-Ve Marks	No.of Qs.	Total marks
Sec – I (Q.N : 1 – 20)	Questions with Single Correct Choice	4	–1	20	80
Sec – II (Q.N : 21 – 30)	Questions with Numerical Answer Type(+ / – Decimal Number)	4	0	10	20
Total				30	100

CHEMISTRY :

Section	Question Type	+Ve Marks	-Ve Marks	No.of Qs.	Total marks
Sec – I (Q.N : 31 – 50)	Questions with Single Correct Choice	4	–1	20	80
Sec – II (Q.N : 51 – 60)	Questions with Numerical Answer Type(+ / – Decimal Number)	4	0	10	20
Total				30	100

MATHEMATICS:

Section	Question Type	+Ve Marks	-Ve Marks	No.of Qs.	Total marks
Sec – I (Q.N : 61 – 80)	Questions with Single Correct Choice	4	–1	20	80
Sec – II (Q.N : 81 – 90)	Questions with Numerical Answer Type(+ / – Decimal Number)	4	0	10	20
Total				30	100

Exam Syllabus

PHYSICS	Basic Mathematics: Trigonometric Ratio's Basic Coordinate Geometry : Distance formulae slope of line, straight line and equations and use in physics Parabola (Quadratic equations) plotting quadratic equations, Ellipse,hyperbola, Rectangular hyperbola, Exponential function [100%] (VECTOR : Introduction to vector quantities and types Addition of vector (Triangle law) Parallelogram law and polygon law Subtraction of vector , Resolution of vectors in 2D and 3D VECTOR : Dot product and its application Cross product & its application MATHEMATICAL PHYSICS : Differentiation, Differentiation as rate measure
CHEMISTRY	Oxidation number, Oxidation and Reduction, Redox reaction disproportionation reaction, Oxidizing & Reducing agents, Calculation of oxidation number by bonding methods. Balancing of redox reactions by ion-electron method, Balancing of redox reactions by Oxidation number method [100%] Introduction to Modern Periodic Table Atomic mass, molar mass, Isotopes, isotones, isobars, Average Atomic mass Mole concept Difference between various states of matter, measurable properties of Gas (Pressure, Temperature and Volume) Ideal Gas Equation and simple numericals related to P, V and T
MATHEMATICS	SET THEORY : Introduction, Sets, Representation of sets, Kinds of Sets, Analysis of two sets : Equal sets, Equivalent sets, Subsets, Intervals as subset of R. Power sets, Universe set, Venn diagram. Operation on sets : Union of sets, intersection of sets, disjoint sets, difference two sets, Complement of a set, Algebra on sets, Practical problems on union and intersection of two sets [100%] Linear Inequalities Wavy Curve Method Logarithmic function and their properties. Determinant evaluation. Cartesian products of sets, Relations and function, Types of Relations. Modulus function and their properties

Rough Work

PHYSICS**Max. Marks: 100****SECTION - I****(SINGLE CORRECT ANSWER TYPE)**

This section contains 20 multiple choice questions. Each question has 4 options (A), (B), (C) and (D) for its answer, out of which **ONLY ONE** option can be correct.

Marking scheme: +4 for correct answer, 0 if not attempted and -1 in all other cases.

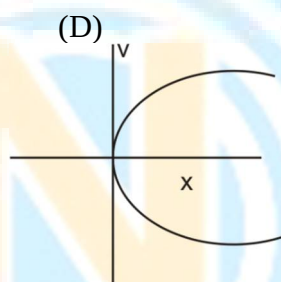
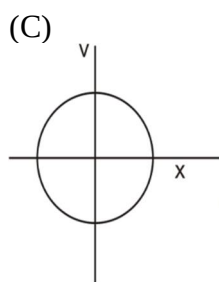
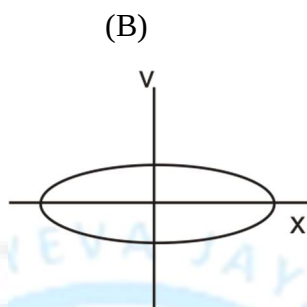
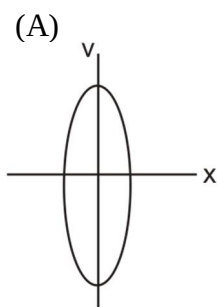
- If two vectors $\vec{P} = \hat{i} + 2m\hat{j} + m\hat{k}$ and $\vec{Q} = 4\hat{i} - 2\hat{j} + m\hat{k}$ are perpendicular to each other. Then, the value of m will be.
(A) -1 (B) 2 (C) 3 (D) 1.
- Statement-I Two forces $(\vec{P} + \vec{Q})$ and $(\vec{P} - \vec{Q})$ where $\vec{P} \perp \vec{Q}$, when act at an angle θ_1 each other, the magnitude of their resultant is $\sqrt{3(P^2 + Q^2)}$, when they act at an angle θ_2 , the magnitude of their resultant becomes $\sqrt{2(P^2 + Q^2)}$. This is possible only when $\theta_1 < \theta_2$.
Statement-II : In the situation given above. $\theta_1 = 60^\circ$ and $\theta_2 = 90^\circ$ In the light of the above statement, choose the most appropriate answer from the options given below:
(A) Statement I is false but Statement II is true.
(B) Both Statement I and Statement II are true.
(C) Both Statement I and Statement II are false.
(D) Statement I is true but Statement II is false.
- Let $\vec{C} = \vec{A} + \vec{B}$, then
(A) $|\vec{C}|$ is always greater than $|\vec{A}|$ (B) It is possible to have $|\vec{C}| < |\vec{A}|$ and $|\vec{C}| < |\vec{B}|$
(C) $|\vec{C}|$ is always equal to $|\vec{A}| + |\vec{B}|$ (D) $|\vec{C}|$ is never equal to $|\vec{A} + \vec{B}|$.
- Let the angle between two non-zero vectors $\vec{A}$ and $\vec{B}$ be 120° and resultant be $\vec{C}$
(A) $|\vec{C}|$ must be equal to $|\vec{A} - \vec{B}|$ (B) $|\vec{C}|$ must be less than $|\vec{A} - \vec{B}|$
(C) $|\vec{C}|$ must be greater than $|\vec{A} - \vec{B}|$ (D) $|\vec{C}|$ may be equal to $|\vec{A} - \vec{B}|$
- $\vec{A} = \hat{i} + \hat{j} - \hat{k}$; $\vec{B} = 2\hat{i} + 3\hat{j} + 5\hat{k}$ angle between $\vec{A}$ and $\vec{B}$ is:
(A) 120° (B) 90° (C) 60° (D) 30°
- A force $\vec{F} = 3\hat{i} + c\hat{j} + 2\hat{k}$ acting on a particle causes a displacement $\vec{d} = -4\hat{i} + 2\hat{j} + 3\hat{k}$. If the work done is 6 joule then the value of 'c' is:
(A) 12 (B) 0 (C) 6 (D) 1.

Rough Work

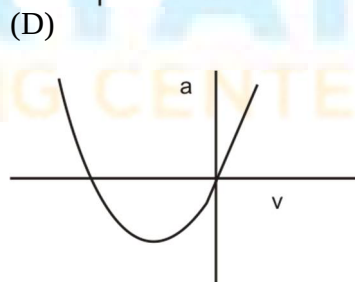
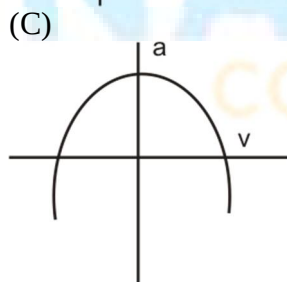
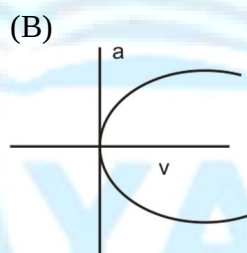
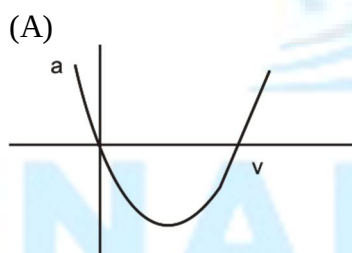
7. Two forces P and Q are in ratio $P:Q = 1:2$. If their resultant is at an angle $\tan^{-1}\left(\frac{\sqrt{3}}{2}\right)$ to vector P, then angle between P and Q is :
- (A) $\tan^{-1}\left(\frac{1}{2}\right)$ (B) 45° (C) 30° (D) 60°
8. If $\hat{a}, \hat{b}, \hat{c}$ are three unit vectors such that $\hat{a} + \hat{b} + \hat{c} = 0$, then $\hat{a} \cdot \hat{b} + \hat{b} \cdot \hat{c} + \hat{c} \cdot \hat{a}$ is equal to:
- (A) -1 (B) 3 (C) 0 (D) $-\frac{3}{2}$
9. If the resultant of three forces $\vec{F}_1 = p\hat{i} + 3\hat{j} - k$, $\vec{F}_2 = -5\hat{i} + \hat{j} + 2\hat{k}$ and $\vec{F}_3 = 6\hat{i} - \hat{k}$ acting on a particle has magnitude equal to 5 units, then the value of $|p|$ is:
- (A) -6 (B) -5 (C) 2 (D) 4.
10. Statement-1 When two non-parallel forces $\vec{F}_1$ and $\vec{F}_2$ act on body, the magnitude of the resultant force acting on the body is less than sum of $|\vec{F}_1|$ and $|\vec{F}_2|$.
- Statement-2 In a triangle, any side is less than the sum of the other two sides.
- (A) Statement-1 is true, statement-2 is true and statement-2 is correct explanation for statement-1.
(B) Statement-1 is true, statement-2 is true and statement-2 is NOT correct explanation for statement-1.
(C) Statement-1 is false, Statement-2 is true.
(D) Statement-1 is true, Statement-2 is false.
11. What is the unit vectors perpendicular to the following vectors $2\hat{i} + 2\hat{j} - \hat{k}$ and $6\hat{i} - 3\hat{j} + 2\hat{k}$?
- (A) $\frac{\hat{i} + 10\hat{j} - 18\hat{k}}{5\sqrt{17}}$ (B) $\frac{\hat{i} - 10\hat{j} + 18\hat{k}}{5\sqrt{17}}$ (C) $\frac{\hat{i} - 10\hat{j} - 18\hat{k}}{5\sqrt{17}}$ (D) $\frac{\hat{i} + 10\hat{j} + 18\hat{k}}{5\sqrt{17}}$
12. Two adjacent sides of a parallelogram are represented by the two vectors $\hat{i} + 2\hat{j} + 3\hat{k}$ and $3\hat{i} - 2\hat{j} + \hat{k}$. What is the area of parallelogram?
- (A) 8 (B) $8\sqrt{3}$ (C) $3\sqrt{8}$ (D) 192.

Rough Work

13. If relation between velocity (v) and position (x) of a particle is given as $\frac{v^2}{9} + \frac{x^2}{4} = 1$ then the graph which represent its equation correctly given by.



14. $a = 3v^2 - 6v$ where a is acceleration and v is velocity then the graph best represented by.



15. If $\sin \theta \approx \theta$ and $(180^\circ = \pi \text{ rad})$ then the value of $\sin(2^\circ)$ in radian will be

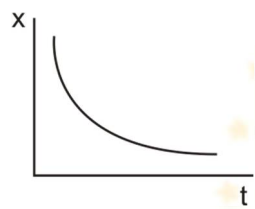
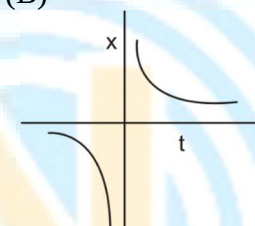
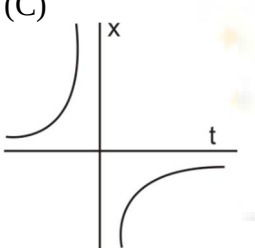
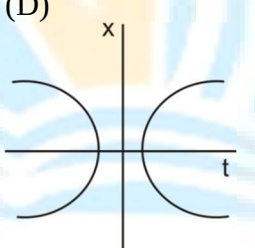
(A) $\frac{\pi}{90}$

(B) $\frac{\pi}{45}$

(C) zero

(D) 2

Rough Work

16. Relation between displacement (S) vs (time) t is given as $S = \log_e(e^{3t^2})$ then particle velocity at t = 2 sec will be in m/sec use $\log_e e = 1$ and velocity $(v) = \frac{d}{dt}(s)$.
- (A) 6 (B) 12 (C) 3 (D) $3t^2$
17. $\frac{d}{dx}(x\sqrt{x}) = ?$
- (A) $\frac{3}{2}\sqrt{x}$ (B) $\frac{2}{3\sqrt{x}}$ (C) $\frac{5}{2}x^{5/2}$ (D) $\frac{1}{2\sqrt{x}}$
18. If $x = \frac{4}{t}$ where x is position in meter and t is time in sec then the graph x vs t is best represented by
- (A) 
- (B) 
- (C) 
- (D) 
19. Let $\vec{A} = (\hat{i} + \hat{j})$ and $\vec{B} = (2\hat{i} - \hat{j})$. The magnitude of a coplanar vector $\vec{C}$ such that $\vec{A} \cdot \vec{C} = \vec{B} \cdot \vec{C} = \vec{A} \cdot \vec{B}$ is given by:
- (A) $\sqrt{\frac{9}{12}}$ (B) $\sqrt{\frac{20}{9}}$ (C) $\sqrt{\frac{5}{9}}$ (D) $\sqrt{\frac{10}{9}}$
20. If $\sin(m\theta) = 2\sin 2\theta \cdot \cos 2\theta$ then m is equal to
- (A) 3 (B) 2 (C) 4 (D) 6

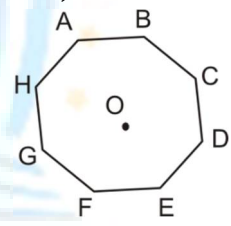
Rough Work

SECTION-II

(Numerical Value Answer Type)

This section contains 10 questions. The answer to each question is a Numerical value. If the Answer is in the decimals, **Mark the nearest Integer only. Have to Answer any 5 only out of 10 questions** and question will be evaluated according to the following marking scheme:

Marking scheme: +4 for correct answer, 0 if not attempted and -1 in all other cases.

21. Vectors $a\hat{i} + b\hat{j} + \hat{k}$ and $2\hat{i} - 3\hat{j} + 4\hat{k}$ are perpendicular to each other when $3a + 2b = 7$, the ratio of a to b is $\frac{x}{2}$. The value of x is_____.
22. If the projection of $2\hat{i} + 4\hat{j} - 2\hat{k}$ on $\hat{i} + 2\hat{j} + a\hat{k}$ is zero. Then, the value of a will be_____.
23. If $\vec{A} = (2\hat{i} + 3\hat{j} - \hat{k})m$ and $\vec{B} = (\hat{i} + 2\hat{j} + 2\hat{k})m$. The magnitude of component of vector $\vec{A}$ along vector $\vec{B}$ will be _____ m.
24. If the projection vector of vector $\vec{A} = \hat{i} + \hat{j} + \hat{k}$ on vector $\vec{B} = \hat{i} + \hat{j}$ is $x\hat{i} + y\hat{j}$ then the value of $x + y$ is equal to _____.
25. A hall has the dimension $10m \times 12m \times 14m$. A fly starting at one corner ends up at a diagonally opposite corner. Then the magnitude of its displacement_____ (in meter).
26. In an octagon ABCDEFGH of equal side, the sum of $\vec{AB} + \vec{AC} + \vec{AD} + \vec{AE} + \vec{AF} + \vec{AG} + \vec{AH} = X \vec{AO}$ then X is _____.

27. If the motion of the particle is governed by the equation $S = 2t^3 - 3t^2 + 2t + 2$ where s is position (S) in meter and time (t) is in sec acceleration of particle at $t = 1\text{sec}$ is x and y respectively the $\frac{y}{x}$ is equal to_____. Where acceleration (a) = $\frac{d}{dt}\left(\frac{d}{dt}(s)\right)$.
28. If a vector $\vec{P}$ make angles α, β and γ respectively with the X, Y and Z axes. Then, $\sin^2 \alpha + \sin^2 \beta + \sin^2 \gamma$ is equal to_____.
29. The sum of two forces $\vec{P}$ and $\vec{Q}$ is $\vec{R}$ such that $|\vec{R}| = |\vec{P}|$. Then the angle between resultant of $2\vec{P} + \vec{Q}$ and $\vec{Q}$ is $90^\circ \times X$. Find the value of X ?
30. A particle moving with velocity v which is varying with time given as $v = 2t - 6$. If graph is plotted between time (t) vs velocity (v) where time is taken along Y axis and velocity is taken along X axis then intercept on Y-axis is_____.

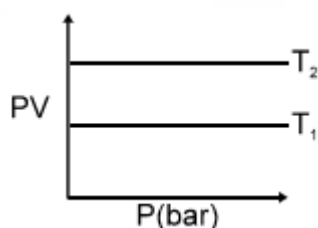
Rough Work

CHEMISTRY**Max.Marks:100****SECTION – I****(SINGLE CORRECT ANSWER TYPE)**

This section contains 20 multiple choice questions. Each question has 4 options (A), (B), (C) and (D) for its answer, out of which **ONLY ONE** option can be correct.

Marking scheme: +4 for correct answer, 0 if not attempted and -1 in all other cases.

31. In the reaction. $P + NaOH \rightarrow PH_3 + NaH_2PO_2$
- (A) P is oxidised only (B) P is reduced only
(C) Na is reduced (D) P is reduced as well as oxidized
32. Which reaction involve neither oxidation nor reduction
- (A) $CrO_4^{2-} \rightarrow Cr_2O_7^{2-}$ (B) $Cr \rightarrow CrCl_3$ (C) $Na \rightarrow Na^+$ (D) $2S_2O_3^{2-} \rightarrow S_4O_6^{2-}$
33. 100 g $MgCO_3$ contains Mg
- (A) 56.5% (B) 28.5% (C) 82% (D) 42%
34. What is the number of atoms in 470 amu of $^{235}_{92}U$?
- (A) 470 (B) $2 \times 6.023 \times 10^{23}$ (C) 2 (D) 5
35. Which of the following has the smallest number of molecules ?
- (A) 22.4×10^3 ml of CO_2 gas (B) 22 g of CO_2 gas
(C) 11.2 litre of CO_2 gas (D) 0.1 mole of CO_2 gas
36. Substance having more number of moles is
- (A) 90 gms of water (B) 112 liter of hydrogen at S.T.P
(C) 24 gm of helium gas (D) 3.01×10^{24} molecules of CO_2 .
37. The number of grams of H_2SO_4 present in 0.25 mole of H_2SO_4 is
- (A) 0.245 (B) 2.45 (C) 24.5 (D) 49.0
38. The product of PV is plotted against pressure at two temperature T_1 and T_2 and the result is shown in figure. What is correct about T_1 and T_2 ?



- (A) $T_1 > T_2$ (B) $T_2 > T_1$ (C) $T_1 = T_2$ (D) $T_1 + T_2 = 1$

Rough Work

39. Atomic radii of Fluorine and Neon in Angstrom units are given by -
(A) 0.72,1.60 (B) 1.60,1.60 (C) 0.72,0.72 (D) None of these.
40. At NTP 1.0 g hydrogen has volume in litre :
(A) 1.12 (B) 22.4 (C) 2.24 (D) 11.2
41. Haemoglobin contains 0.335% iron (at. Wt. of Fe = 56). The molecular weight of haemoglobin is 1.67×10^4 . Find the number of iron atoms in the haemoglobin.
(A) 2 (B) 3 (C) 4 (D) 1
42. An element exists in nature in two isotopic forms: X^{30} (90%) and X^{32} (10%). What is the average atomic mass of element?
(A) 30.2amu (B) 60.4amu (C) 15.1amu (D) 30.2gram .
43. Which of the following samples contains the largest number of atoms ?
(A) 1 g of Ni(s) (B) 1 g of Ca(s) (C) 1 g of $N_2(g)$ (D) 1 g of B(s)
44. Which of the following is a redox reaction?
(A) $NaCl + KNO_3 \rightarrow KNO_3 + NaCl$ (B) $CaC_2O_4 + 2HCl \rightarrow CaCl_2 + H_2C_2O_4$
(C) $Mg(OH)_2 + 2NH_4Cl \rightarrow MgCl_2 + 2NH_4OH$ (D) $Zn + 2AgCN \rightarrow 2Ag + Zn(CN)_2$.
45. Which of the following is a disproportionation reaction?
(A) $Cu_2O + 2H^+ \rightarrow Cu + Cu^{2+} + H_2O$ (B) $2CrO_4^{2-} + 2H^+ \rightarrow Cr_2O_7^{2-} + H_2O$
(C) $CaCO_3 + 2H^+ \rightarrow Ca^{2+} + H_2O + CO_2$ (D) $Cr_2O_7^{2-} + 2\overset{\ominus}{O}H \rightarrow 2CrO_4^{2-} + H_2O$.
46. At fixed temperature and 600 mm pressure, the density of a gas is 42. At the same temperature and 700 mm pressure, what is the density of the gas?
(A) 29 (B) 25 (C) 49 (D) 55.
47. At which of the following four conditions will the density of nitrogen be the largest?
(A) STP (B) 273 K and 2 atm.
(C) 546 K and 1 atm (D) 546 K and 2 atm.
48. Assuming the atomic weight of a metal M to be 56, find the empirical formula of its oxide containing 70.00% of M.
(A) M_2O_3 (B) MO_3 (C) MO_2 (D) M_3O_2 .

Rough Work

49. Which of the following contains maximum number of atoms ?
 (A) 10 g of CaCO_3 (B) 7.1 g of Cl_2 (C) 18 g of $\text{C}_6\text{H}_{12}\text{O}_6$ (D) 5.6 g of N_2
50. The increasing order of atomic radii of the following Group 13 element is
 (A) $\text{Al} < \text{Ga} < \text{In} < \text{Tl}$ (B) $\text{Ga} < \text{Al} < \text{In} < \text{Tl}$
 (C) $\text{Al} < \text{In} < \text{Ga} < \text{Tl}$ (D) $\text{Al} < \text{Ga} < \text{Tl} < \text{In}$.

SECTION-II

(Integer Value Answer Type)

This section contains 10 questions. The answer to each question is a Numerical value. If the Answer is in the decimals. **Have to Answer any 5 only out of 10 questions** and question will be evaluated according to the following marking scheme:

Marking scheme: +4 for correct answer, 0 if not attempted and -1 in all other cases.

51. Calculate the mass of Na_2CO_3 which will have the same number of molecules as contained in 12.3 g of $\text{MgSO}_4 \cdot 7\text{H}_2\text{O}$ (nearest integer) ?
52. The number of species among the following, having inert gas configuration is.....
 $\text{K}^+, \text{Ca}^{2+}, \text{S}^-, \text{S}^{2-}, \text{Br}^-, \text{Se}^{2-}, \text{H}^+, \text{H}^-, \text{Mn}^{2+}$
53. The number of atom in 8 g of sodium is $x \times 10^{23}$. The value of x is.....(Nearest integer)
54. Number of hydrogen atoms per molecule of a hydrocarbon A having 85.8% carbon is ____
 (Given: Molar mass of A = 84 g mol^{-1}).
55. A gas occupies a volume of 2.5L at $9 \times 10^5 \text{Nm}^{-2}$. The additional pressure required to decrease the volume of the gas to 1.5L, keeping temperature constant ____ $\times 10^5 \text{Nm}^{-2}$.
56. A sample of nitrogen occupies a volume of 320cm^3 at STP. Calculate its volume at 546.3 K and 0.5 bar pressure.
57. What is the mass of carbon present in 0.5 mole of $\text{K}_4[\text{Fe}(\text{CN})_6]$?
58. The difference in the oxidation number of two type of sulphur atom $\text{Na}_2\text{S}_4\text{O}_6$ is ____.
59. What is the oxidation no of Cr in CrO_5
60. $2\text{MnO}_4^- + b\text{C}_2\text{O}_4^{2-} + c\text{H}^+ \rightarrow x\text{Mn}^{2+} + y\text{CO}_2 + z\text{H}_2\text{O}$. If the above equation is balanced with integer coefficients the value of c is ____.

Rough Work

MATHEMATICS**Max.Marks:100****SECTION – I****(SINGLE CORRECT ANSWER TYPE)**

This section contains 10 multiple choice questions. Each question has 4 options (A), (B), (C) and (D) for its answer, out of which **ONLY ONE** option can be correct.

Marking scheme: +4 for correct answer, 0 if not attempted and -1 in all other cases.

61. If P, Q and R are subsets of a set A , then $R \times (P^c \cup Q^c)^c =$
- (A) $(R \times P) \cap (R \times Q)$ (B) $(R \times Q) \cap (R^c \times P)$
- (C) $(R \times P) \cup (R \times Q)$ (D) None of these
62. Let A and B be two unequal sets. Then
- (A) $A \cup B \subseteq A \cap B$ (B) $A \cap B \subseteq A \cup B$
- (C) $A \cap B = A \cup B$ (D) None of these
63. Let $A = \{(x, y) : y = e^x, x \in \mathbb{R}\}$, $B = \{(x, y) : y = e^{-x}, x \in \mathbb{R}\}$ Then
- (A) $A \cap B = \phi$ (B) $A \cap B \neq \phi$
- (C) $A \cup B = \mathbb{R}^2$ (D) None of these
64. Given that, x is a real number satisfying $\frac{5x^2 - 26x + 5}{3x^2 - 10x + 3} < 0$, then
- (A) $x < \frac{1}{5}$ (B) $\frac{1}{5} < x < 3$
- (C) $x > 5$ (D) $\frac{1}{5} < x < \frac{1}{3}$ or $3 < x < 5$
65. The solution of the equation $\log_{101} \log_7 (\sqrt{x+7} + \sqrt{x}) = 0$ is
- (A) 3 (B) 7 (C) 9 (D) 49
66. The set of all real x satisfying the inequality $\frac{3-|x|}{4-|x|} \geq 0$, is
- (A) $[-3, 3] \cup (-\infty, -4) \cup (4, \infty)$ (B) $(-\infty, -4) \cup (4, \infty)$
- (C) $(-\infty, -3) \cup (4, \infty)$ (D) $(-\infty, -3) \cup (3, \infty)$

Rough Work

67. The solution of $\frac{6x}{4x-1} < \frac{1}{2}$ is
- (A) $x < \frac{-1}{8}$ (B) $\frac{-1}{8} < x < \frac{1}{4}$
- (C) $x < \frac{-1}{8}$ and $x > \frac{1}{4}$ (D) $x > \frac{1}{8}$
68. Sum of values of x and y satisfying $\log_x(\log_3(\log_x y)) = 0$ and $\log_y 27 = 1$ is :
- (A) 27 (B) 30 (C) 33 (D) 36
69. The difference of roots of the equation $(\log_{27} x^3)^2 = \log_{27} x^6$ is :
- (A) $\frac{2}{3}$ (B) 1 (C) 9 (D) 8
70. If $T_r = \frac{1}{\log_2 r^4}$ (where $r \in N$), then the value of $\sum_{r=1}^4 T_r$ is :
- (A) 3 (B) 4 (C) 5 (D) 10
71. The domain of function $f(x) = \sqrt{\frac{x^2-7x+10}{x^2-5x+6}}$ is
- (A) $(-\infty, 3] \cup [5, \infty)$ (B) $(-\infty, 3) \cup (5, \infty)$
- (C) $(-\infty, 3) \cup [5, \infty) - \{2\}$ (D) $(-\infty, 3] \cup (5, \infty) - \{2\}$
72. Number of integral values of x which satisfies the inequality $\frac{(x+6)^2(x+5)(x+1)^2}{(x-3)} \leq 0$, is
- (A) 10 (B) 6 (C) 8 (D) 9
73. Complete solution of inequality $||3x - 9| + 2| > 2$ is
- (A) $(-\infty, \infty)$ (B) $\{3\}$ (C) $R - \{3\}$ (D) ϕ
74. If $\begin{vmatrix} 1 & 2 \\ 3 & 4 \end{vmatrix} = \begin{vmatrix} 5 & x \\ 1 & 2 \end{vmatrix}$, then x is given by
- (A) 1 (B) 5 (C) 10 (D) 12
75. If $\left| \frac{3x+4}{3} \right| = 7$, then x is
- (A) $\frac{17}{3}, \frac{25}{3}$ (B) $-\frac{17}{3}, \frac{25}{3}$ (C) $\frac{17}{3}, -\frac{25}{3}$ (D) $-\frac{17}{3}, -\frac{25}{3}$

Rough Work

76. Values of x satisfying the equality $|x^2 + 8x + 7| = |x^2 + 4x + 4| + |4x + 3|$ for $x \in R$ are
 (A) $(-2, \infty)$ (B) $(-\infty, \frac{-3}{4}]$ (C) $[\frac{-3}{4}, \infty) \cup \{-2\}$ (D) None of these
77. If $\log_6 16 = k$, then $\log_{18} 24 =$
 (A) $\frac{2(k+2)}{k-8}$ (B) $\frac{2(k+2)}{8-k}$ (C) $\frac{k+8}{k}$ (D) $\frac{k}{k+8}$
78. The value of x satisfying the equation $4^{\log_9 3} + 9^{\log_2 4} = 10^{\log_x 83}$ is equal to
 (A) 9 (B) 4 (C) 10 (D) 100
79. Let A and B be two finite sets such that $n(A) = 20, n(B) = 28$ and $n(A \cup B) = 36$, find $n(A \cap B)$.
 (A) 8 (B) 10 (C) 12 (D) 14
80. If $n(A - B) = 18, n(A \cup B) = 70$ and $n(A \cap B) = 25$, then find $n(B)$.
 (A) 32 (B) 42 (C) 52 (D) 62

SECTION-II (Numerical Value Answer Type)

This section contains 5 questions. The answer to each question is a Numerical value. If the Answer is in the decimals, **Mark the nearest Integer only.** Question will be evaluated according to the following marking scheme:

Marking scheme: +4 for correct answer, 0 if not attempted and -1 in all other cases.

81. If $x = \prod_{n=1}^{2000} n$, then the value of the expression,

$$\frac{1}{\frac{1}{\log_2 x} + \frac{1}{\log_3 x} + \dots + \frac{1}{\log_{2000} x}} \text{ is}$$

82. Find number of negative integers satisfying $\left| \frac{x^2 - 5x + 4}{x^2 - 4} \right| \leq 1$

83. Find number of single digit prime numbers satisfying

$$\frac{(x-2)^{100}(2+x)^{101}(x-1)^{10}}{x^{41}(x+1)^{39}(4-x)^{11}} \leq 0$$

84. Number of integral values of x that satisfies

$$x^{x\sqrt{x}} = (x\sqrt{x})^x \text{ is}$$

Rough Work

85. Find the modulus of the sum of all solutions to $(x^2 + 3x + 1)^{x^2+2x-8} = 1$.
86. If $\begin{vmatrix} 1 & 4 & 0 \\ 0 & 1 & 2 \\ -1 & 2 & k \end{vmatrix} = 0$, then k is equal to
87. If $\frac{x^2 + 2x + 7}{2x + 3} < 6, x \in R$, find the largest possible integral solution.
88. If the least integral value of x that satisfies $\frac{x-5}{x^2+5x-14} > 0$ is α , find $|\alpha|$.
89. If $n(A \cap B^c) = 5, n(B \cap A^c) = 6, n(A \cap B) = 4$ then the value of $n(A \cup B)$ is
90. A market research group conducted a survey of 1000 consumers and reported that 720 consumers like product A and 450 consumers like product B , the least number that must have liked both products is

Rough Work